Advanced Vehicle State Monitoring: Evaluating Moving Horizon Estimators and Unscented Kalman Filter

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***Abstract*—Active safety systems must be used to manipulate the dynamics of autonomous vehicles to ensure safety. To this end, accurate vehicle information, such as the longitudinal and lateral velocities, is crucial. Measuring these states, however, can be expensive, and the measurements can be polluted by noise. The available solutions often resort to Bayesian filters, such as the Kalman filter, but can be vulnerable and erroneous when the underlying assumptions do not hold. With its clear merits in handling nonlinearities and uncertainties, moving horizon estimation (MHE) can potentially solve the problem and is thus studied for vehicle state estimation. This paper designs an unscented Kalman filter, standard MHE, modified MHE, and recursive least squares MHE to estimate critical vehicle states, respectively. All the estimators are formulated based upon a highly nonlinear vehicle model that is shown to be locally observable. The convergence rate, accuracy, and robustness of the four estimation algorithms are comprehensively characterized and compared under three different driving maneuvres. For MHE-based algorithms, the effects of horizon length and optimization techniques on the computational efficiency and accuracy are also investigated.**

***Index Terms*—Moving horizon estimation, vehicle state estimation, kalman filter, nonlinear observability.**

I. Introduction

AUTONOMOUS vehicles (AVs) have been drawing the attention of both traditional car manufacturers, e.g., Volvo and Audi, and technology companies like Google and Baidu. With the advance of AV technology, active safety systems such as electronic stability control (ESC) and anti-lock braking system (ABS) are playing an increasingly important role, especially in safety-critical situations. For example, in 2015 ESC saved 1,949 lives in passenger vehicle crashes in the USA, while there were still 2,272 fatalities resulting from these accidents [1]. To ensure the best possible performance from such life-saving systems, an accurate knowledge of vehicle states is important. However, states like longitudinal and lateral velocities cannot be measured directly with cost-effective sensors, leading to a demand for reliable online estimation algorithms.

Kinematics-based techniques, typically utilising various sensor measurements and the transformation among sensor positions, have been widely used to estimate desired vehicle states [2]–[4]. To describe where the measuring devices are installed, four different frames or their combinations are often employed, including the earth frame for which the global positioning system (GPS) is employed, the vehicle body frame for inertial measurement unit (IMU) installation, the vehicle frame for describing the vehicle motion, and the wheel frame [5]. In this class of approaches, the vehicle sideslip angle is usually estimated by directly integrating noisy/biased IMU measurements, subtracting the heading angle from measured GPS course angle, or the combination of the two. Although being easy to implement, these schemes are vulnerable to sensor bias and faults, and potential GPS outages. Moreover, they are limited by low update frequencies or a high cost of GPS devices [6].

To compensate for the limitation of kinematics-based techniques, dynamics-based approaches have been proposed for state estimation, requiring vehicle and tyre models together with physical parameters. These approaches are often realised by fusing different vehicle/tyre models and real-time state estimation techniques. For instance, based on a bicycle model of the vehicle and a linear tyre model, Anderson and Bevly [7] adopted the Kalman filter (KF) to detect sideslip angle and yaw rate. By using the KF and recursive least squares embedded similar system models, Nam *et al.* [8] estimated the sideslip and roll angles of electric vehicles using lateral tire force sensors. These linear model-based algorithms are superior in computational efficiency but suffer from large model uncertainties because the small angle assumption underpinning the bicycle model cannot hold when the vehicle/tyre behaves in some highly nonlinear regions [9].

To maintain the estimation performance in all possible operating regions, complicated nonlinear vehicle models and tyre models, such as the Pacejka tyre model [10], have been pursued

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in estimator design. Based upon these nonlinear models, extended Kalman filters (EKFs) [11], [12] and unscented Kalman filters (UKFs) [13]–[15] have been widely exploited to gauge different vehicle states. To in some degree address the modelling error, Katriniok and Abel [16] introduced two additional adaptation states to scale longitudinal and lateral tire forces within the EKF framework. For the same purpose, Liu *et al.* [17] proposed the minimum model error criterion combined with the EKF. Techniques and applications of adaptive UKFs for state estimation have been well documented in [18], [19]. Although these KF-type estimators have shown promising results, relying on linearisation or unscented transformation to approximate the nonlinear system will potentially introduce additional modelling errors, thus lowering the estimator performance such as accuracy and convergence rate. Equipped with Monte Carlo algorithms, the Rao-Blackwellized particle filter (PF) have been applied to jointly estimate key variables in active safety systems, such as the longitudinal velocity, roll angle, and wheel slip for all four wheels [20]. In addition to the computational burden, this PF estimator turned to be not sufficiently robust in dealing with unknown persistent disturbances and noise.

As different to the above estimation methods that merely involve the current measurement, the moving horizon estimation (MHE) algorithm derives state estimates by solving an optimisation problem in terms of a window of the most recent measurements at each time step [21], [22]. By incorporating various sequential measurements for state estimation, MHE tends to be more adaptive and robust in response to uncertainties stemmed from modelling, parametrisation, and measurements. Moreover, nonlinear vehicle/tyre models can be formulated naturally into the MHE framework, thus maintaining a high model fidelity for state prediction. In light of these two aspects, the MHE has emerged in the field of vehicle state/parameter estimation, e.g., [23]–[25]. These salient upsides, however, are achieved at the cost of extensive computational resources that are required to solve the nonlinear optimisation problem in real-time, especially when multiple state variables exist and a large horizon is considered. Therefore, the implementation of MHE for advanced monitoring and management of vehicle dynamics is subject to competing objectives, such as system complexity and estimation accuracy, and well balancing different objectives is of great importance.

From the above discussions, it is clear that a comprehensively comparative study of different MHE and KF-type estimation algorithms is desired in the context of vehicle state estimation. This research problem is meticulously addressed in the present work for estimating vital vehicle states. Three contributions are made to distinguish this work from available vehicle estimation efforts. First, with a nonlinear system model synthesised from the double-track vehicle model and simplified Pacejka tyre model, an observability analysis is conducted and shows that the system model is locally observable. Then, the standard MHE, a modified MHE (mMHE) adopted from [26], the recursive least squares MHE (RLS-MHE), and the UKF are designed successively to monitor the longitudinal and lateral velocities, yaw rate, yaw angle, and position of the vehicle. Their convergence rates, accuracy and robustness are characterised and compared under three different driving manoeuvres. In addition, the effects of horizon length and optimisation techniques are evaluated in terms of the estimation performance and computational efficiency.

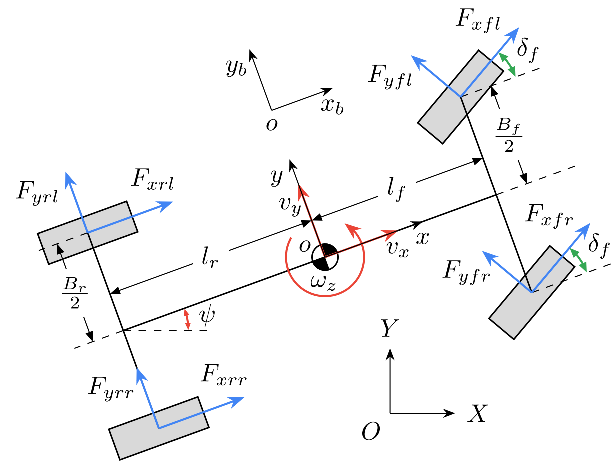


Fig. 1. Illustration of the planar vehicle model considering longitudinal, lateral, and yaw movements.

II. Vehicle Dynamics Modeling

This section details the system model that governs vehicle motion, tyre forces, and vertical forces for estimation application. This is followed by introduction of a more accurate yet complex model that will be used to mimic the system plant and evaluate the estimation-oriented model in simulation studies. Both models are formulated in the frames in accordance with the ISO forward-leftward-upward (FLU) definition [5].

A. Model for Estimation

As illustrated in Fig. 1, the planar vehicle model synthesised from [17] and [27] is used to describe the system dynamics that consider longitudinal, lateral and yaw movements. Three frames, namely the earth frame XOY, the vehicle body frame $x_b o y_b$, and the vehicle frame xoy, are combined to define the state variables. By using the symbols defined in Table I, the planar vehicle model can be readily mathematically formulated with the governing equations as

$$\begin{aligned} m\dot{v_x} &= (F_{xfl} + F_{xfr})\cos\delta_f - (F_{yfl} + F_{yfr})\sin\delta_f \\ &\quad + F_{xrl} + F_{xrr} + m v_y \omega_z \end{aligned} \tag{1}$$

$$\begin{aligned} m\dot{v_y} &= (F_{xfl} + F_{xfr})\sin\delta_f + (F_{yfl} + F_{yfr})\cos\delta_f \\ &\quad + F_{yrl} + F_{yrr} - m v_x \omega_z \end{aligned} \tag{2}$$

$$\begin{aligned} I_z\dot{\omega_z} &= \tfrac{B_f}{2}((F_{xfr} - F_{xfl})\cos\delta_f + (F_{yfl} - F_{yfr})\sin\delta_f) \\ &\quad + \tfrac{B_r}{2}(F_{xrr} - F_{xrl}) + l_f((F_{xfl} + F_{xfr})\sin\delta_f \\ &\quad + (F_{yfl} + F_{yfr})\cos\delta_f) - l_r(F_{yrl} + F_{yrr}) \end{aligned} \tag{3}$$

$$\dot{\psi} = \omega_z \tag{4}$$

$$\dot{X} = v_x \cos\psi - v_y \sin\psi \tag{5}$$

$$\dot{Y} = v_x \sin\psi + v_y \cos\psi \tag{6}$$

TABLE I
GLOBAL NOMENCLATURE AND NOTATION

$\mathcal{A}$	A set in which the elements denote the front left, front right, rear left and rear right wheels, respectively, $= \{fl, fr, rl, rr\}$.
v_x	Longitudinal velocity at centre of gravity (CoG) $(\mathrm{m/s})$.
v_y	Lateral velocity at CoG $(\mathrm{m/s})$.
ω_z	Yaw rate around CoG $(\mathrm{rad/s})$.
ψ	Yaw angle (rad).
X	Longitudinal position (m).
Y	Lateral position (m).
a_x	Longitudinal acceleration $(\mathrm{m/s^2})$.
a_y	Lateral acceleration $(\mathrm{m/s^2})$.
δ_f	Steering angle, mean of two front wheels (rad).
ω_i	Angular velocity of the wheel $(\mathrm{rad/s})$ $(i \in \mathcal{A})$.
κ_i	Tyre slip ratio $(i \in \mathcal{A})$.
α_i	Tyre slip angle (rad) $(i \in \mathcal{A})$.
F_{zi}	Vertical tyre force (N) $(i \in \mathcal{A})$.
F_{xi}	Longitudinal tyre force (N) $(i \in \mathcal{A})$.
F_{yi}	Lateral tyre force (N) $(i \in \mathcal{A})$.
m	Vehicle mass (kg).
I_z	Vehicle yaw inertia $(\mathrm{kg \cdot m^2})$.
g	Gravitational acceleration $(\mathrm{m/s^2})$.
B_f	Front track width (m).
B_r	Rear track width (m).
l_f	Distance from CoG to front axle (m).
l_r	Distance from CoG to rear axle (m).
h_g	Height of CoG (m).
r_e	Tyre radius (m).

where (1)–(3) convert the dynamics of the vehicle body frame to those of the vehicle frame, and (4)–(6) transform the motion in the vehicle frame into that in the earth frame.

To calculate the longitudinal and lateral tyre forces, F_{xi} and F_{yi}, required in (1)–(3), $G_{x\alpha i}$ and $G_{y\kappa i}$ represent the effect of the lateral/longitudinal slip on the longitudinal/lateral tyre forces; $B_{x\alpha i}$, $B_{y\kappa i}$, $C_{x\alpha i}$ and $C_{y\kappa i}$ are the corresponding coefficients determining that effect. These coefficients rely on the tyre parameters r_{bx1i}, r_{bx2i}, r_{cx1i}, r_{by1i}, r_{by2i}, r_{by3i} and r_{cy1i}. According to [28], the longitudinal and lateral tyre forces can be derived through a simplified Pacejka tyre model

$$F_{xi} = G_{x\alpha i} \cdot F_{x0i},\ F_{yi} = G_{y\kappa i} \cdot F_{y0i} \tag{7a}$$

$$G_{x\alpha i} = \cos(C_{x\alpha i} \arctan(B_{x\alpha i} \tan \alpha i)) \tag{7b}$$

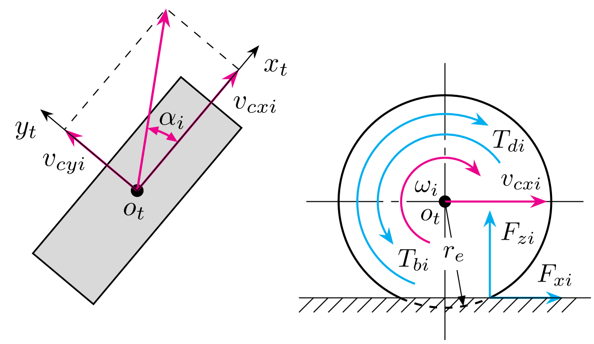
$$G_{y\kappa i} = \cos(C_{y\kappa i} \arctan(B_{y\kappa i} \kappa i)) \tag{7c}$$

$$B_{x\alpha i} = r_{bx1i} \cos(\arctan(r_{bx2i} \kappa)) \tag{7d}$$

$$B_{y\kappa i} = r_{by1i} \cos(\arctan(r_{by2i}(\tan \alpha_i - r_{by3i}))) \tag{7e}$$

$$C_{x\alpha i} = r_{cx1i},\ C_{y\kappa i} = r_{cy1i} \tag{7f}$$

where F_{x0i} and F_{y0i} $(i \in \mathcal{A})$ denote the longitudinal and lateral tyre forces, respectively, under the pure slip condition. F_{x0i} and F_{y0i} can be calculated from the Pacejka tyre equations [10, eqs. (4.E9) and (4.E19)]. For simplicity in this calculation, the effect



(a) (b)

Fig. 2. Wheel frame and rotational motion. (a) and (b) show the top and side views, respectively.

of the camber angle and the pneumatic pressure is usually not considered, and the friction coefficient is not scaled. The two aspects are adopted in this work to develop a vehicle model for estimator design.

The expressions for the tyre slip ratio κ_i and slip angle α_i are given as [10]

$$\kappa_i = \frac{\omega_i r_e - v_{cxi}}{v_{cxi}},\ \alpha_i = \tan \frac{v_{cyi}}{v_{cxi}} \tag{8}$$

where v_{cxi} and v_{cyi} stand for the longitudinal and lateral velocities at the wheel centre, respectively, in the wheel frame $x_t o_t y_t$, as shown in Fig. 2(a). The velocities in this wheel frame are related to those in the vehicle frame by [17]

$$\begin{bmatrix} v_{xfl} \\ v_{xfr} \\ v_{xrl} \\ v_{xrr} \end{bmatrix} = \begin{bmatrix} v_x - B_f \omega_z/2 \\ v_x + B_f \omega_z/2 \\ v_x - B_r \omega_z/2 \\ v_x + B_r \omega_z/2 \end{bmatrix} \quad \begin{bmatrix} v_{yfl} \\ v_{yfr} \\ v_{yrl} \\ v_{yrr} \end{bmatrix} = \begin{bmatrix} v_y + l_f \omega_z \\ v_y + l_f \omega_z \\ v_y - l_r \omega_z \\ v_y - l_r \omega_z \end{bmatrix} \tag{9a}$$

$$\begin{bmatrix} v_{cxj} \\ v_{cyj} \end{bmatrix} = \begin{bmatrix} \cos \delta_f & \sin \delta_f \\ -\sin \delta_f & \cos \delta_f \end{bmatrix} \begin{bmatrix} v_{xj} \\ v_{yj} \end{bmatrix}, \quad j \in \{fl, fr\} \tag{9b}$$

$$\begin{bmatrix} v_{cxj} \\ v_{cyj} \end{bmatrix} = \begin{bmatrix} v_{xj} \\ v_{yj} \end{bmatrix}, \quad j \in \{rl, rr\} \tag{9c}$$

where v_{xi} and v_{yi} are the longitudinal and lateral velocities (at the wheel centre), respectively, in the vehicle frame xoy.

To determine the longitudinal and lateral tyre forces defined by (7), the vertical tyre forces are also required in addition to (8). Without considering the coupling effect between the pitch and roll dynamics, the steady-state vertical forces can be approximated by [11]

$$F_{zfl} = \frac{l_r mg}{2(l_f + l_r)} - \frac{h_g m a_x}{2(l_f + l_r)} - \frac{l_r h_g m a_y}{B_f(l_f + l_r)} \tag{10a}$$

$$F_{zfr} = \frac{l_r mg}{2(l_f + l_r)} - \frac{h_g m a_x}{2(l_f + l_r)} + \frac{l_r h_g m a_y}{B_f(l_f + l_r)} \tag{10b}$$

$$F_{zrl} = \frac{l_f mg}{2(l_f + l_r)} + \frac{h_g m a_x}{2(l_f + l_r)} - \frac{l_f h_g m a_y}{B_r(l_f + l_r)} \tag{10c}$$

$$F_{zrr} = \frac{l_f mg}{2(l_f + l_r)} + \frac{h_g m a_x}{2(l_f + l_r)} + \frac{l_f h_g m a_y}{B_r(l_f + l_r)}. \tag{10d}$$

The measurement equations that correspond to the longitudinal and lateral accelerations, a_x and a_y, are given as

$$a_x = \frac{1}{m}((F_{xfl} + F_{xfr})\cos\delta_f - (F_{yfl} + F_{yfr})\sin\delta_f + F_{xrl} + F_{xrr}) \tag{11}$$

$$a_y = \frac{1}{m}((F_{xfl} + F_{xfr})\sin\delta_f + (F_{yfl} + F_{yfr})\cos\delta_f + F_{yrl} + F_{yrr}). \tag{12}$$

By combining (1)–(12), a nonlinear model describing the vehicle dynamics has been derived. The measured longitudinal and lateral accelerations are denoted by a_x^m and a_y^m, respectively. For simplicity of presentation, the system state vector x, the input vector u and the output vector y are defined as

$$x = \begin{bmatrix} v_x & v_y & \omega_z & \psi & X & Y \end{bmatrix}^T \tag{13a}$$

$$u = \begin{bmatrix} \delta_f & \omega_{fl} & \omega_{fr} & \omega_{rl} & \omega_{rr} & a_x^m & a_y^m \end{bmatrix}^T \tag{13b}$$

$$y = \begin{bmatrix} a_x & a_y & \omega_z & X & Y \end{bmatrix}^T. \tag{13c}$$

As can be seen from (10), the vehicle acceleration is used to calculate the vertical tyre forces which are then used to determine the longitudinal and lateral tyre forces. Besides, the acceleration is also selected as the output variable, thus resulting in an implicit output function. This issue was tackled in [15] by calculating the vertical forces through vertical stiffness and suspension displacement and in [16] by introducing a correlated covariance term in the designed estimator. In this study, this problem is addressed outside the estimator framework by using the measurements a_x^m and a_y^m instead of unknown a_x and a_y to compute the vertical forces in (10).

With the definition in (13), the system model is represented in a compact state-space form

$$\dot{x}(t) = f_c(x(t), u(t)) \tag{14a}$$

$$y(t) = h_c(x(t), u(t)) \tag{14b}$$

where $f_c(\cdot)$ and $h_c(\cdot)$ are the continuous-time state and output functions, respectively, with the explicit form in (1)–(12).

For the sake of digital implementation, the above continuous-time system (14) is discretised using the forward Euler method with sampling time T_s. The obtained discrete-time system is given by

$$x_{k+1} = f(x_k, u_k) \tag{15a}$$

$$y_k = h(x_k, u_k) \tag{15b}$$

where $f(\cdot)$ and $h(\cdot)$ are the discrete-time state and output functions derived from $f_c(\cdot)$ and $h_c(\cdot)$, respectively.

B. Vehicle Plant

A model with seven degrees of freedom (DoF) is utilised to simulate the vehicle plant and to generate the measurement signals. Besides the longitudinal, lateral and yaw motions, the rotational motion of the four wheels is also taken into consideration, as shown in Fig. 2(b). The wheel dynamics are described by

$$I_{wi}\dot{\omega}_i = T_{di} - T_{bi} - r_e F_{xi} - C_{ri} r_e F_{zi} \tag{16}$$

where I_{wi}, T_{di}, T_{bi} and C_{ri} $(i \in \mathcal{A})$ are the rotational inertia, driving torque, braking torque and rolling resistance at the wheel, respectively.

In the seven-DoF vehicle plant model, the air drag force is included in the longitudinal dynamics, which results in an equation that is slightly different from (1). The new longitudinal dynamics are governed by

$$m\dot{v}_x = (F_{xfl} + F_{xfr})\cos\delta_f - (F_{yfl} + F_{yfr})\sin\delta_f + F_{xrl} + F_{xrr} + m v_y \omega_z - C_d A_f \frac{D_a v_x^2}{2} \tag{17}$$

where C_d, A_f and D_a are the drag coefficient, frontal area and air density, respectively.

An additional characteristic of the seven-DoF vehicle plant model is that the complete Pacejka tyre equations [10, eqs. (4.E1)–(4.E67)] are used to determine the longitudinal and lateral tyre forces in the combined slip condition. Similar to the case when calculating F_{x0i} and F_{y0i}, the camber, pneumatic pressure change and friction coefficient scaling are not taken into account here. The entire vehicle plant model is comprised of (2)–(6), (8)–(12), (16)–(17) and the Pacejka tyre model.

C. Model Evaluation

This section validates the simplified model in Section II-A against the plant model in Section II-B. This is performed under three manoeuvres, including a modified double lane change (DLC) [27], an ISO DLC [29] and a circular driving manoeuvre. In the modified DLC manoeuvre, the vehicle is controlled to track the reference path and to have a constant longitudinal velocity. In the ISO DLC manoeuvre, the vehicle first accelerates from a given initial velocity and then runs at an approximately constant speed; meanwhile, the vehicle is controlled to follow the predefined path. The circular manoeuvre represents a very harsh driving condition under which the vehicle accelerates and tries to maintain a reference yaw rate, $\omega_{zref} = v_x/R$, with R being the radius of the desired circular path. In these three driving scenarios, vehicle measurements are sampled at 0.01 s. The model parameters are taken from the software package CarSim [30]. Specifically, the tyre parameters follow the module CS_car225_60R18. The vehicle parameters are set up as: $I_z = 3234$ kg $\cdot$ m^2, $B_f = 1.6$ m, $B_r = 1.6$ m, $l_f = 1.4$ m, $l_r = 1.65$ m, $r_e = 0.3636$ m, $C_d = 0.3$, $A_f = 2.8$ m^2, $C_{ri} = 0.018$ $(i \in \mathcal{A})$, $m = 1759.14$ kg, $h_g = 0.53$ m, and $I_{wi} = 1.636$ kg $\cdot$ m^2 $(i \in \mathcal{A})$. The gravitational acceleration is $g = 9.81$ m/s^2 and the air density is considered as $D_a = 1.206$ kg/m^2.

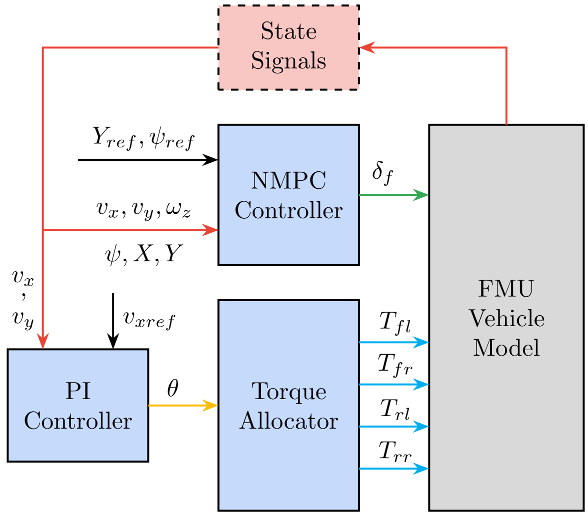


Fig. 3. Vehicle dynamics co-simulation using an FMU.

The vehicle plant model is first built in Dymola and then is exported into MATLAB/Simulink in the form of an FMU (a functional mock-up unit) for co-simulation. The block diagram for simulating vehicle dynamics is given in Fig. 3, where the vehicle control system mainly consists of a nonlinear model predictive controller (NMPC) and a proportional-integral (PI) controller. The NMPC utilises the vehicle signals v_x, v_y, ω_z, ψ, X and Y to determine the current state of the vehicle, and takes Y_{ref} and ψ_{ref} as references, as conducted in [27], [31]. By tracking these reference signals, the front steering angle δ_f is obtained. The PI controller is designed to track the reference longitudinal velocity v_{xref}, and a commanded pedal position signal θ is the output from this controller. The pedal position signal is utilised by the torque allocator to distribute the torque equally to the four individual wheels. The torque command together with the steering angle is employed for vehicle control. Note that the controllers are not tuned further since the control performance is not the focus of this study. Additionally, the variable-step ODE45 (Dormand-Prince) solver [32] is used within MATLAB/Simulink during the simulations.

The results from the model evaluation for the modified DLC manoeuvre are displayed in Fig. 4. As can be seen, the simplified model is able to well match the plant model. In particular, the root mean square (RMS) errors of v_x, v_y and ψ are 0.041 m/s, 0.050 m/s and 0.002 rad, respectively; the maximum errors are 0.104 m/s, 0.204 m/s and 0.004 rad, respectively. Similar results can be observed for the ISO DLC manoeuvre (figure not shown for brevity).

The model outputs for the circular driving manoeuvre are shown in Fig. 5. The system model cannot well capture the plant dynamics except those of v_x, and the modelling errors for v_y, ϕ and w_z in general are increasing with time, leading to not satisfactory model prediction results. The obvious modelling errors are mainly caused from the simplification of tyre model (7). Specifically, due to the neglect of some combined slip components, this simplified tyre model cannot well reproduce the true tyre behaviour during the circular manoeuvre where a heavy interaction between the lateral and longitudinal slip exists. This circular driving manoeuvre is intentionally designed and will also be used in Section IV to test the capability of designed estimators in compensating for large model-plant mismatch.

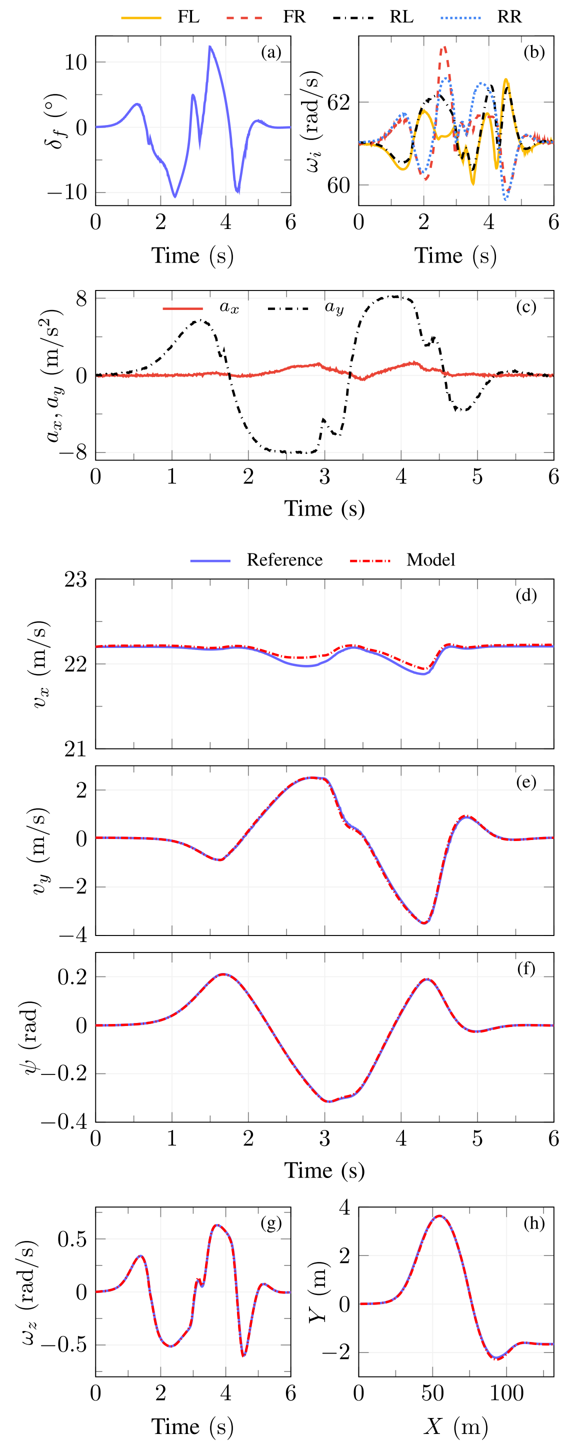


Fig. 4. Modified DLC manoeuvre input signals and model output. (a)–(c) show the input signals, where "FL", "FR", "RL", and "RR" denote the front left, front right, rear left, and rear right wheels, respectively. (d)–(h) present the model output, where "Reference" and "Model" denote the output from the model described in Section II-B and II-A, respectively.

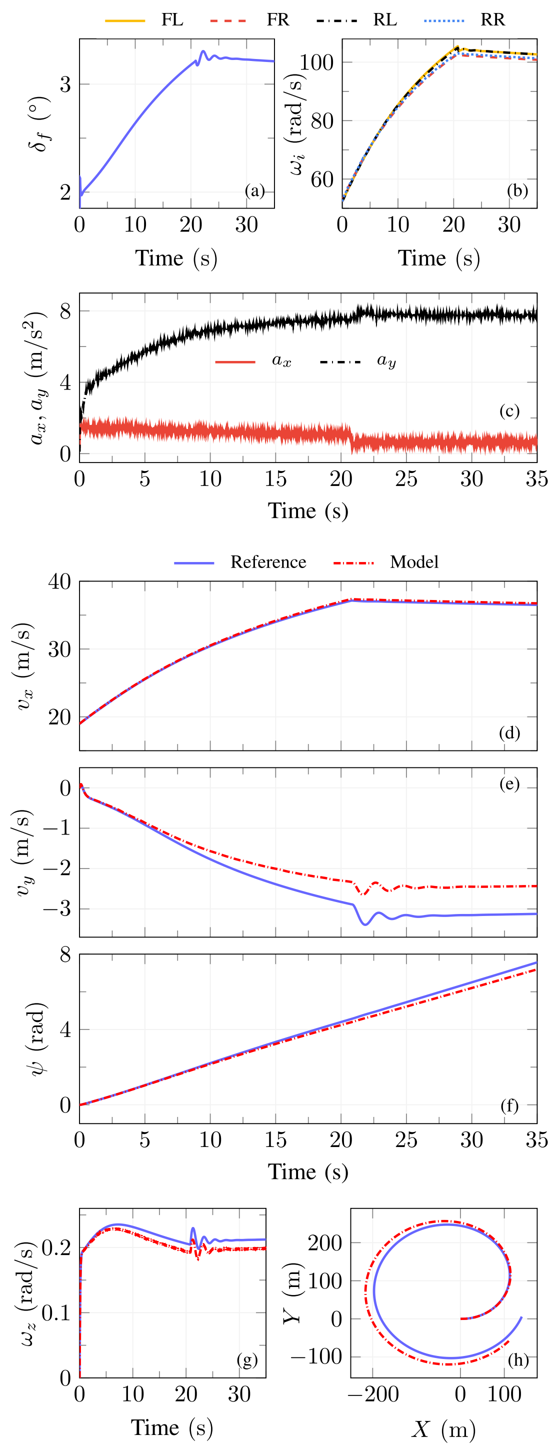


Fig. 5. Circular manoeuvre input signals and model output. (a)–(c) show the input signals. (d)–(h) present the model output.

III. Estimator Design

In this section, the observability of the nonlinear system (14) described in Section II-A is analysed. This is followed by the design of three MHE-type estimators, including a standard one and two variants, as well as the UKF, based on the vehicle model (15).

A. Observability Analysis

The observability of a linear time-invariant system can be determined by examining the observability Gramian. For a nonlinear system, the observability, however, cannot be evaluated directly from the linearised system. The nonlinear system, (14), is said to be locally observable at the current operating point (x_0, u_0) if the following *observability matrix* $\mathcal{O}$ has the same rank as the dimension of the state vector denoted n_x [33]:

$$\begin{aligned}\mathcal{O} &= \begin{bmatrix} \frac{\partial L_f^0 h}{\partial x} & \frac{\partial L_f^1 h}{\partial x} & \cdots & \frac{\partial L_f^{n_x-1} h}{\partial x}\end{bmatrix}^T_{(x_0,u_0)} \\ &:= \begin{bmatrix} J_{L_f^0 h} & J_{L_f^1 h} & \cdots & J_{L_f^{n_x-1} h}\end{bmatrix}^T_{(x_0,u_0)} \end{aligned} \tag{18}$$

where $L_f^i h$ is the i-th order *Lie derivative*, and $J_{L_f^i h}$ is the corresponding Jacobian matrix.

Combining (1)–(12), the first two components in $\mathcal{O}$ are derived as

$$J_{L_f^0 h} = \begin{bmatrix} J_{0,11} & J_{0,12} & J_{0,13} & 0 & 0 & 0 \\ J_{0,21} & J_{0,22} & J_{0,23} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{19a}$$

$$J_{L_f^1 h} = \begin{bmatrix} J_{1,11} & J_{1,12} & J_{1,13} & 0 & 0 & 0 \\ J_{1,21} & J_{1,22} & J_{1,23} & 0 & 0 & 0 \\ J_{1,31} & J_{1,32} & J_{1,33} & 0 & 0 & 0 \\ \cos\psi & -\sin\psi & 0 & J_{1,44} & 0 & 0 \\ \sin\psi & \cos\psi & 0 & J_{1,54} & 0 & 0 \end{bmatrix} \tag{19b}$$

where $J_{0,ij}$ and $J_{1,ij}$ are elements of the corresponding Jacobian matrices, and $J_{1,44} = -v_x \sin\psi - v_y \cos\psi$ and $J_{1,54} = v_x \cos\psi - v_y \sin\psi$.

As can be seen from (19a), the maximum possible rank of $J_{L_f^0 h}$ is 5. Since this maximum value is very close to the rank requirement of 6, it is possible to omit some orders of the *Lie derivative* in the *observability matrix*. From the equations of $J_{1,44}$ and $J_{1,54}$, it can be inferred that $J_{1,44}$ and $J_{1,54}$ will not be zero at the same time if v_x or v_y is nonzero. This is true when the vehicle is moving. Moreover, only the fourth column of $L_f^0 h$ is filled with constant zeros. Therefore, it is viable to use only the first two terms in $\mathcal{O}$ to examine whether the system is locally observable or not. For this purpose, a new *observability matrix*, $\mathcal{O}_2$, is used for observability evaluation

$$\mathcal{O}_2 = \begin{bmatrix} \frac{\partial L_f^0 h}{\partial x} \\ \frac{\partial L_f^1 h}{\partial x} \end{bmatrix}_{(x_0,u_0)} = \begin{bmatrix} J_{L_f^0 h} \\ J_{L_f^1 h} \end{bmatrix}_{(x_0,u_0)}. \tag{20}$$

By numerically calculating the rank of $\mathcal{O}_2$ at each time step, the system is validated to be locally observable since the rank plot is a horizontal line of rank 6.

B. Estimator Design

1) MHE: MHE can be formed in two frameworks: a deterministic one and a stochastic one [34]. For MHE in the first framework, the component corresponding to process noise should receive a small weighting (large covariance) if one has more faith in measurements, and a large weighting (small covariance) if one has less faith in measurements. In the stochastic framework, the covariances can be treated in the same way as in the EKF formulation. Given a time window of measurements, the MHE problem can be formulated as [35]

$$\min_{\boldsymbol{x},\boldsymbol{w},\boldsymbol{v}} \left\|x_{k-N} - x_{k-N|k-N-1}\right\|^2_{P^{-1}_{k-N|k-N-1}} + \sum_{i=k-N}^{k-1} \left\|w_i\right\|^2_{Q^{-1}} + \sum_{i=k-N}^{k} \left\|v_i\right\|^2_{R^{-1}} \tag{21a}$$

$$\text{s.t. } x_{i+1} = f(x_i, u_i) + w_i \tag{21b}$$

$$y_i = h(x_i, u_i) + v_i \tag{21c}$$

where $\boldsymbol{x} = [x_{k-N}, \ldots, x_k]$, $\boldsymbol{w} = [w_{k-N}, \ldots, w_{k-1}]$ and $\boldsymbol{v} = [v_{k-N}, \ldots, v_k]$ are the state variable, process noise and measurement noise sequences, respectively, and N is the length of the estimation window (horizon length). The process and measurement noise covariances are represented by Q and R, respectively. A state estimate with the subscript $i|k$ denotes the estimate at time step i with measurements until time step k. With this definition, $x_{k-N|k-N-1}$ and $P_{k-N|k-N-1}$ are called the *a priori* state estimate and the corresponding error covariance at time step $k-N$, respectively. Similarly, $x_{k-N|k-N}$ and $P_{k-N|k-N}$ are the *a posteriori* state estimate and error covariance at time step $k-N$, respectively. It is intuitive to see that a small weighting, i.e. Q^{-1}, should be given to the second component in (21a) if a large covariance Q is used, meaning that little trust is put on the model, and vice versa. This is also the reason for $P^{-1}_{k-N|k-N-1}$ and R^{-1} being used in the cost function.

Ideally, all the past measurements can be used to estimate the states at each time step, but this will cause overly demanding computational burden. By contrast, only $N+1$ measurements are considered in the MHE formulation in (21). To summarise the missing past information, an arrival cost term, the first term in (21), is required. The arrival cost can be updated by EKF and UKF-based approaches [36], [37], or filtering/smoothing update strategy [21].

When EKF is employed for updating the arrival cost, the EKF formulation needs to be modified to prevent the repeated usage of some measurements [36, eqs. (40)–(45)]. In the present study, for simpler implementation the standard EKF formulation has been utilised. The measurement correction step, (22a)–(22c), and prediction step, (22d)–(22e), are given as [38]

$$K_{k-N} = P_{k-N|k-N-1} C^T_{k-N|k} (R + C_{k-N|k} P_{k-N|k-N-1} C^T_{k-N|k})^{-1} \tag{22a}$$

$$x_{k-N|k-N} = x_{k-N|k-N-1} + K_{k-N}(y_{k-N} - h(x_{k-N|k}, u_{k-N})) \tag{22b}$$

$$P_{k-N|k-N} = (I - K_{k-N} C_{k-N|k}) P_{k-N|k-N-1} \tag{22c}$$

$$x_{k-N+1|k-N} = f(x_{k-N|k}, u_{k-N}) \tag{22d}$$

$$P_{k-N+1|k-N} = A_{k-N|k} P_{k-N|k-N} A^T_{k-N|k} + Q \tag{22e}$$

where K_k is the Kalman gain. The Jacobian matrices $A_{k-N|k}$ and $C_{k-N|k}$, which are linearised around the smoothed estimate $x_{k-N|k}$, are defined as

$$A_{k-N|k} = \left.\frac{\partial f}{\partial x}\right|_{x_{k-N|k}} \quad C_{k-N|k} = \left.\frac{\partial h}{\partial x}\right|_{x_{k-N|k}}. \tag{23}$$

The above updating strategy for the arrival cost is developed for the case when $t > N$. On other occasions, MHE can be implemented either by forging faked measurements or by truncating the horizon [39]. In the present study, the latter procedure has been adopted, in which the MHE horizon length is initialised at zero. Then the horizon is increased gradually to N, after which the MHE estimator operates with a fixed horizon.

2) mMHE: The MHE (21)-(23) can be tailored to not estimate the process noise for reduced computational overhead. A possible way is to treat the process model as deterministic by following [26]. As a result, the process noise term w is removed from the model and objective function, leading to the mMHE estimator being formulated as

$$\min_{\boldsymbol{x},\boldsymbol{v}} \left\|x_{k-N} - x_{k-N|k-N-1}\right\|^2_{P^{-1}_{k-N|k-N-1}} + \sum_{i=k-N}^{k} \left\|v_i\right\|^2_{R^{-1}} \tag{24a}$$

$$\text{s.t. } x_{i+1} = f(x_i, u_i) \tag{24b}$$

$$y_i = h(x_i, u_i) + v_i. \tag{24c}$$

Note that even though the uncertainty in the process is not directly included in (24), the effect of process noise is partly taken into account in the arrival cost term. In other words, by approximating the arrival cost with (22), the process noise is partly handled by the covariance Q in (22e).

3) RLS-MHE: The RLS-MHE estimator is a further simplified version of the MHE estimator. In addition to the process noise, the arrival cost term in the objective function is also discarded. The resulted estimator is given by

$$\min_{\boldsymbol{v}} \sum_{i=k-N}^{k} \left\|v_i\right\|^2_{R^{-1}} \tag{25a}$$

$$\text{s.t. } x_{i+1} = f(x_i, u_i) \tag{25b}$$

$$y_i = h(x_i, u_i) + v_i. \tag{25c}$$

4) UKF: Similar to EKF, UKF also consists of a time update step and a measurement correction step. However, instead of approximating the nonlinear system with first-order transformation of mean and covariance like EKF, unscented transformation (UT) is used by UKF, which produces a third-order accuracy. By utilising UT a small number of sigma points that have the desired mean and covariance are selected. Each of the sigma points is then propagated with the nonlinear function, yielding transformed sigma points. These transformed points are subsequently used to obtain an estimate of the mean and covariance resulting from the nonlinear transformation. In the present study, the UKF time and measurement update steps are derived by following [38, eqs. (14.56)–(14.67)], and their explicit formulation is ignored here for brevity.

IV. Results and Discussion

The four estimation algorithms formulated in the previous section are now implemented and comprehensively compared. Trajectories of the system input u are generated from the manoeuvres mentioned earlier, i.e., the modified DLC, ISO DLC and circular manoeuvres. In each driving scenario, the vehicle plant model described in Section II-B is used to simulate the true vehicle and to generate measurement signals. To test the robustness of each estimator, the measurements are polluted by adding artificial noise of normal distribution with mean 0 and standard deviation $0.05\ \mathrm{m/s^2}$, $0.03\ \mathrm{m/s^2}$, $0.005\ \mathrm{rad/s}$, $0.1\ \mathrm{m}$, and $0.1\ \mathrm{m}$ for the longitudinal acceleration, lateral acceleration, yaw rate, longitudinal position, and lateral position, respectively.

The MHE and variants are all implemented in MPCTools [39], which is an interface to CasADi [40]. CasADi provides the first and second-order derivatives to the optimisation package IPOPT [41] and the solver MA27 [42]. Within IPOPT and MA27, the nonlinear optimisation problems are solved to full convergence with default solver settings, except for a time limit 20 s for the RLS-MHE. Furthermore, all simulations are performed in MATLAB R2016b running on a laptop computer configured with an Intel i7-6820HQ processor.

For a fair comparison, the tuning parameters of the four estimators are chosen to be the same. The prior guess covariance P_0, the process and measurement noise covariances Q and R, and the horizon length N (for MHE-type estimators) are set as

$$P_0 = \mathrm{diag}([0.005 \quad 0.005 \quad 0.01 \quad 0.1 \quad 0.1 \quad 0.1])^2 \tag{26a}$$

$$Q = \mathrm{diag}([0.02 \quad 0.05 \quad 0.01 \quad 0.03 \quad 0.1 \quad 0.1])^2 \tag{26b}$$

$$R = \mathrm{diag}([0.01 \quad 0.03 \quad 0.001 \quad 0.01 \quad 0.01])^2 \tag{26c}$$

$$N = 10. \tag{26d}$$

In addition, to test their robustness and convergence rate, the estimators are initialised with poor prior guesses. Specifically, a $15\,\%$ error for v_x and a fixed deviation for the other states are subtracted from the true initial state x_0. The initial guess used in the modified DLC manoeuvre is $x_0 - [3.33 \quad 1 \quad 0.3 \quad 0.5 \quad 1 \quad 1]^T$, while the initial guesses in both the ISO DLC and circular manoeuvres are $x_0 - [2.85 \quad 1 \quad 0.3 \quad 0.5 \quad 1 \quad 1]^T$.

A. Convergence and Accuracy

Prior to presenting the results, the time to converge denoted t_{cvg} is used as an evaluation index. As unknown noise and disturbance exist persistently, the convergence of estimation errors to the origin may not be possible. To facilitate the comparison, all estimators are considered to have converged if at time step k

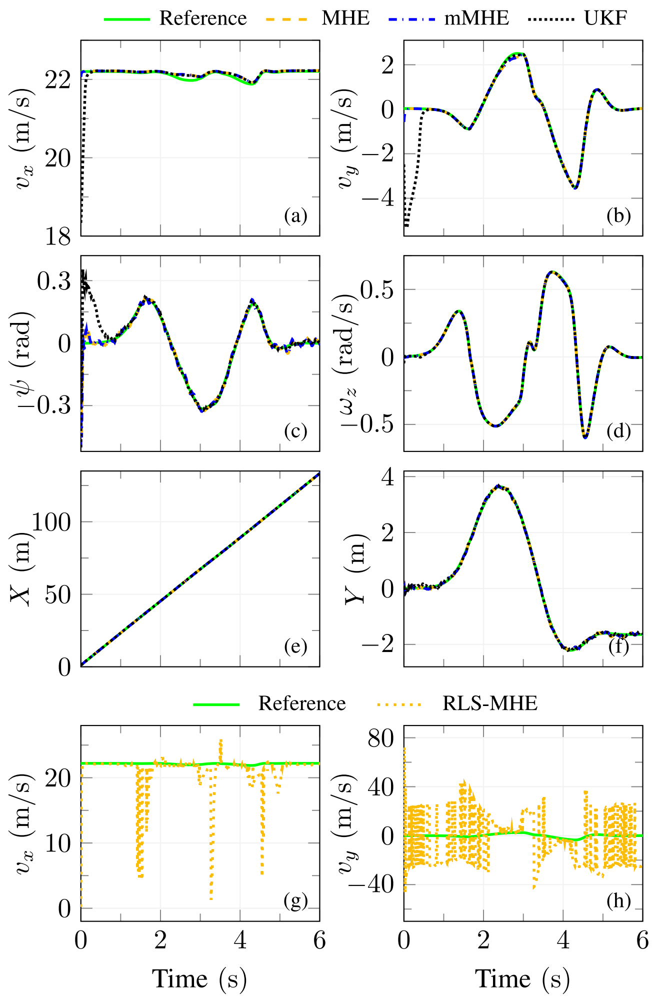


Fig. 6. Modified DLC manoeuvre estimation. (a)–(f) show a comparison without the RLS-MHE estimator. (g) and (h) present only the velocity estimates from the RLS-MHE estimator.

the true state x_k and the estimated state $x_{k|k}$ satisfy

$$|x_k - x_{k|k}| \leq \begin{bmatrix} 0.2 & 0.2 & 0.1 & 0.04 & 0.3 & 0.3 \end{bmatrix}^T. \tag{27}$$

The values selected on the right-hand side of (27) are reasonable for this study with specifications and parameters defined before, though they are not rigorous, nor sufficiently general for all vehicles under various driving situations.

The vehicle state estimates for the modified DLC and ISO DLC manoeuvres are revealed in Figs. 6–7 and Tables II–III. In the modified DLC manoeuvre (see Table II and Fig. 6), the MHE converges to the true values in only 0.21 seconds, which is more than two times faster than the corresponding convergence of the UKF. Besides, the v_y estimate from the MHE starts to converge once the estimator is initialised, while that from the UKF initially deviates abruptly from the poor prior guess ($-1\ \mathrm{m/s}$). This can be explained as follows. As mentioned earlier, the MHE is initialised with a truncated horizon. This means that the number of measurements included in the MHE formulation, (21)–(23), increases with an increasing horizon length when $t \leq N$. As a result, the arrival cost term, or poor prior guess, in (21) plays a

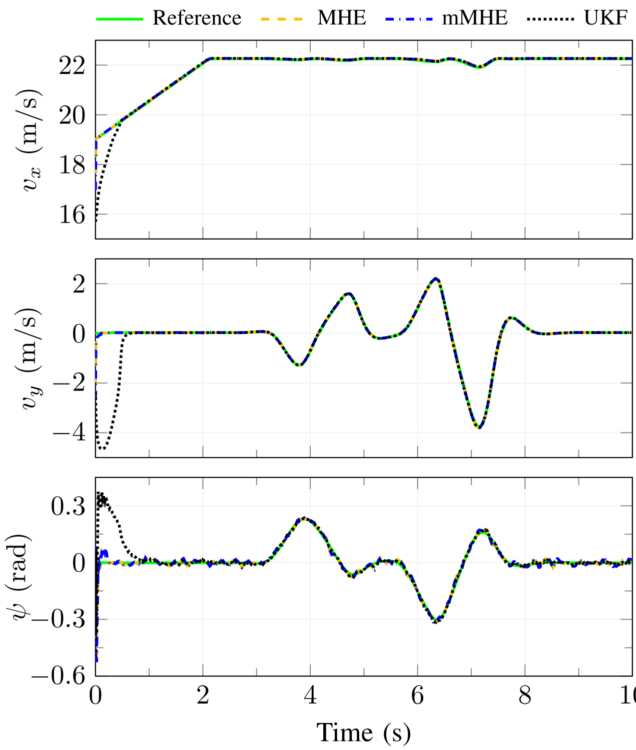


Fig. 7. ISO DLC manoeuvre estimation.

TABLE II
MODIFIED DLC MANOEUVRE ESTIMATION

Estimator	v_x (m/s)		v_y (m/s)		ψ (rad)		t_{cvg} (s)
	e_{max}	e_{rms}	e_{max}	e_{rms}	e_{max}	e_{rms}	
MHE	0.137	0.049	0.606	0.064	0.500	0.034	0.21
mMHE	0.137	0.049	0.606	0.063	0.500	0.034	0.21
RLS-MHE	22.00	2.994	72.66	22.30	55.60	7.813	> 6
UKF	3.864	0.345	5.368	1.033	0.500	0.067	0.58

[a]Concerning Table II and the following tables, t_{cvg} is calculated in such a way that all the estimated states converged to the true vehicle states.
[b]Some estimated states from the RLS-MHE estimator deviate too far from the ground truth, which makes the estimation meaningless, and therefore those values are represented by $-$.

TABLE III
ISO DLC MANOEUVRE ESTIMATION

Estimator	v_x (m/s)		v_y (m/s)		ψ (rad)		t_{cvg} (s)
	e_{max}	e_{rms}	e_{max}	e_{rms}	e_{max}	e_{rms}	
MHE	2.034	0.067	2.645	0.088	0.526	0.029	0.22
mMHE	2.034	0.068	2.645	0.088	0.526	0.029	0.22
RLS-MHE	–	–	–	–	98.99	17.70	> 10
UKF	3.319	0.318	4.683	0.848	0.500	0.065	0.68

less important role in the solution-finding process, and therefore the MHE is less affected by the inaccurate initialisation. The mMHE has similar performance to that of the MHE, because the model mismatch under such a manoeuvre is small so that the hypothesis of a deterministic model underlying (24) holds.

Quantitatively, the maximum and RMS errors of v_x, v_y and ψ estimates from the MHE are 0.137 m/s and 0.049 m/s, 0.606 m/s and 0.064 m/s, and 0.500 rad and 0.034 rad, respectively. These errors are similar to those from the mMHE but are at least $49\,\%$ smaller (except for the e_{max} of ψ) than those from the UKF. The RLS-MHE behaves much worse than the UKF. Specifically, it does not converge in the tested scenarios, as indicated in Fig. 6(g)–(h) and Table II. A similar behaviour was also reported in [26] for a chemical process monitoring and in [43] for battery state estimation, where large deviations and abrupt changes were observed from the RLS-MHE. The divergence of the RLS-MHE is mainly due to the missing the *a priori* estimate in its objective function, (25a), and the only tunable parameter is R. These factors combined could lead to the system uncertainty being addressed less effectively by the RLS-MHE than by the other estimators. Since the results from the RLS-MHE deviate considerably from the true values, they are only partly shown in Fig. 6(g)–(h) and in the tables following them. Moreover, the RLS-MHE estimator is not analysed further hereafter. For the ISO DLC manoeuvre (see Table III and Fig. 7), similar results to those presented above can be observed.

From Fig. 6, it is also observed that the yaw rate and vehicle position estimates from the MHE, mMHE and UKF estimators are barely affected by the poor prior guess. One possible explanation for this is that a higher trust is given to the measurements although these states exist both in the state vector, (13a), and the output vector, (13c). In fact, even the RLS-MHE estimator gives reliable estimates for these states. For this reason, these states are not shown in Table II or in the following tables and figures.

B. Robustness Test

To test the robustness of the designed estimators, three scenarios are considered, including increased initial deviation, modelling error and external disturbance.

1) Large Initial Deviation: A simple sensitivity analysis is conducted to test the robustness of the estimators in dealing with even larger initial errors. In the ISO DLC manoeuvre, even when the initial v_y or ψ deviation is increased to double its original value, the results from all the estimators do not change much with regard to the convergence time, but there are slightly larger maximum and RMS errors. However, if the v_x deviation is increased from $15\,\%$ to $25\,\%$, it takes the UKF more than 7 seconds to converge, but this increase barely affects the convergence of the MHE and mMHE. As mentioned earlier, other states are measured directly and the measurements are given a high level of trust, so the initial deviation of those variables has little effect on the convergence. The above observations indicate that the estimators are more sensitive to v_x deviation, and the UKF estimator is more sensitive to larger initialisation errors than the MHE and mMHE estimators.

2) Large Modeling Errors: As observed earlier, the modelling errors increase dramatically with time in the circular manoeuvre. For example, the v_y error at 30 s becomes as large as $-22.4\,\%$ of the true value. The modelling errors are to a high degree compensated for by the UKF estimator, as can be seen from Fig. 8 and Table IV. The errors are restrained further by the MHE

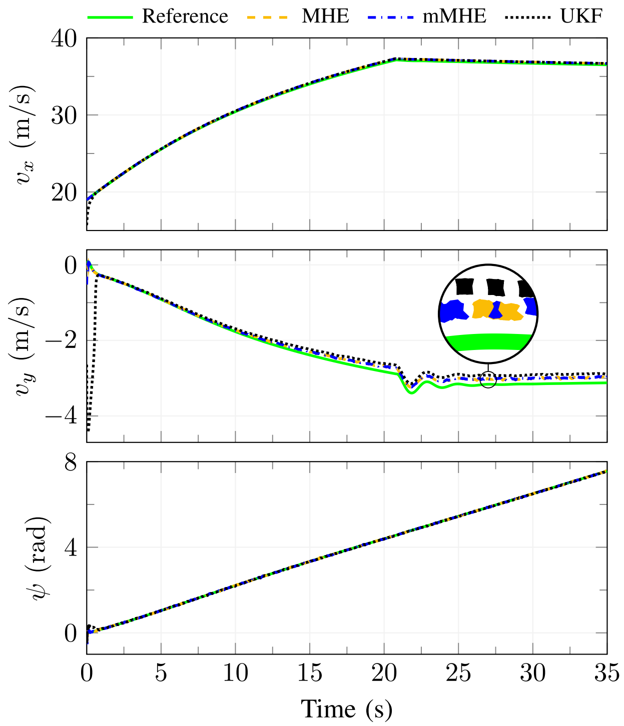


Fig. 8. Circular manoeuvre estimation.

TABLE IV
CIRCULAR MANOEUVRE ESTIMATION

Estimator	v_x (m/s)		v_y (m/s)		ψ (rad)		t_{cvg} (s)
	e_{max}	e_{rms}	e_{max}	e_{rms}	e_{max}	e_{rms}	
MHE	0.191	0.134	0.523	0.110	0.526	0.017	0.22
mMHE	0.190	0.135	0.523	0.112	0.526	0.018	0.22
RLS-MHE	–	64.70	–	–	327.3	14.09	> 35
UKF	3.308	0.198	4.517	0.478	0.500	0.033	0.75

and mMHE estimators, as indicated in Fig. 8, especially when the errors become more obvious ($t > 20$ s). Specially, the v_y error from the MHE and mMHE at 30 s is -0.157 m/s, which is 37 % better than the results obtained with the UKF. The MHE and mMHE are also considerably better than the others when evaluating on the basis of the maximum and RMS errors.

3) External Disturbance: In this test the system is assumed to be perturbed by a measurement disturbance $y_{dstb} = [5\,5\,5\,0\,0]^T$ for one sample at the time instant $t = 4.5$ s. As can be seen from Fig. 9, in general the MHE has a markedly smaller estimation error than all the other estimators, especially for the v_y and ψ estimates. The mMHE, on the other hand, is even worse than the UKF. This can be explained from an optimisation point of view. As discussed previously, after eliminating the equality constraints, there are $(N+1) \times n_x$ optimisation variables in the MHE, but only n_x such variables in the mMHE. Likewise, the EKF is similar to the MHE with a horizon length of 1 [21]. This means that the MHE has the largest number of optimisation variables, while the mMHE has much fewer such variables (undisturbed components), with the result that the mMHE is easily contaminated by disturbances. The behaviour of the

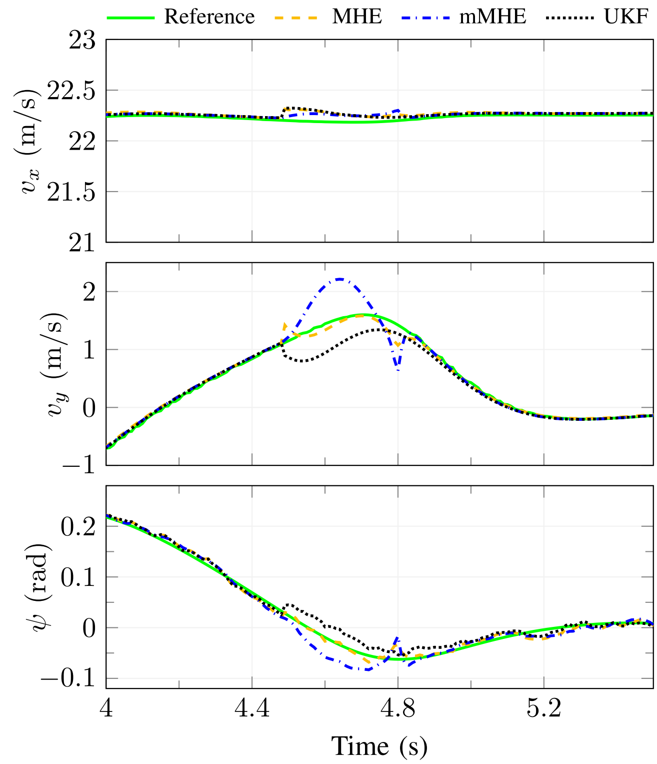


Fig. 9. ISO DLC manoeuvre estimation with measurement disturbance.

UKF, in this regard, can be explained analogously to that of the EKF. In this case study, the horizon length of the MHE and mMHE is increased to 30 to lower the negative effect of the perturbed measurements further. This measure is taken because, as stated in [21], using a short horizon may cause unexpected estimation errors if the system is contaminated by an unmodelled disturbance.

C. Effect of Horizon Length and Optimization Techniques

The effect of horizon length on the performance of the MHE and mMHE estimators in the circular manoeuvre is displayed in Fig. 10. As can be seen, the accuracy of the MHE does not apparently improve with the increasing horizon length. This is mainly because large modelling errors exist in the circular manoeuvre and compromise the optimisation process although more measurements are available. Still, the MHE in general yields considerably better results than the mMHE as the horizon length increases. Specifically, the RMS error of v_y resulting from the MHE is 37% smaller than that from the mMHE, with the horizon length being 60. This is again due to the fact that the process model is assumed to be a perfect one in the mMHE without completely considering the modelling errors.

An additional consequence of having a large horizon length is the increased computational burden in the optimisation process. The change of computation time, which is evaluated using the solver output from the MPCTools, with the horizon length is illustrated in Fig. 10(d). In particular, it takes the MHE 6.47 ms and the mMHE 6.29 ms to solve the optimisation problem with a horizon length of 10. Moreover, their computation time grows

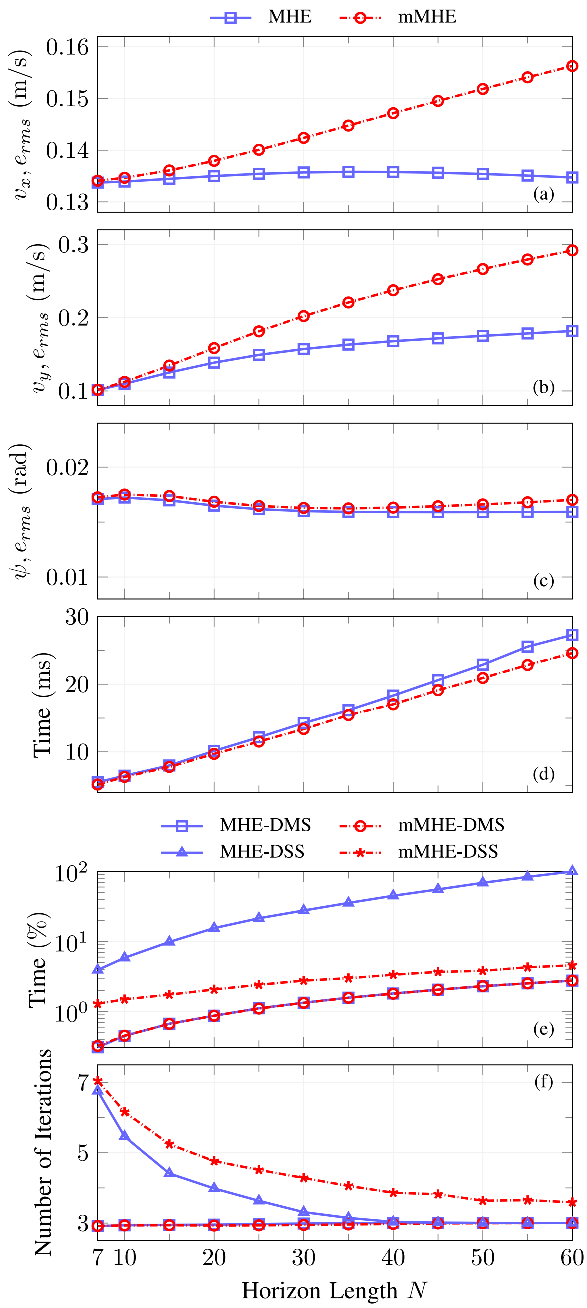


Fig. 10. Effect of horizon length and optimisation technique in the circular manoeuvre. (a)–(c) show the RMS errors with the direct multiple shooting. (d) shows the average computation time per optimisation step with the direct multiple shooting. (e) and (f) show the average computation time per optimisation step (the y tick value has been scaled with respect to the maximum value), and the average number of iterations per optimisation step, both with the CasADi Opti. “DMS” and “DSS” denote direct multiple shooting and direct single shooting, respectively.

dramatically with an increasing horizon length. However, the computation of the MHE does not vary much from that of the mMHE, which is contrary to the results obtained in [26] and [43]. This is because a different optimisation technique, direct multiple shooting [44], is used in the present study. With the direct multiple shooting method, the equality constraints (21b)–(21c) in the MHE and (24b)–(24c) in the mMHE are handled as defect constraints to ensure continuity [45]. Furthermore, x, w and v are all treated as optimisation variables. As a result, there are $(N+1)\times(n_x+n_y)+N\times n_x$ optimisation variables and $N\times n_x+(N+1)\times n_y$ equality constraints in the MHE, where n_y is the number of output variables. Similarly, there are $(N+1)\times(n_x+n_y)$ optimisation variables and $N\times n_x+(N+1)\times n_y$ equality constraints in the mMHE.

In [26] and [43], however, the single shooting method was utilised to solve the optimisation problem, where the equality constraints were eliminated from the problem formulation. Consequently, there were $(N+1)\times n_x$ and n_x optimisation variables in the MHE and the mMHE, respectively. Moreover, no equality constraints were considered there. It is apparent that the complexity of the optimisation problem does not differ between the MHE and mMHE estimators in the present study as much as it differs between them in [26] and [43], thus resulting in a similar computation time for the estimators in the present study.

To investigate further the effect of different optimisation techniques, a second study of the computational time is carried out where both estimators are implemented using the CasADi Opti functionality to be more comparable. Specifically, the MHE and mMHE are both solved by the direct multiple shooting and direct single shooting methods, respectively. As can be seen from Fig. 10(e), the choice of the shooting techniques significantly affects the computational efficiency of the two algorithms. Firstly, the computation time of the two estimators with the direct multiple shooting is substantially less than that with the direct single shooting, which is particularly obvious for the MHE. Secondly, with the direct multiple shooting the computation of the MHE again grows similarly to that of the mMHE as the horizon length increases, but the computational cost of the MHE increases incredibly faster than that of the mMHE when the direct single shooting is used.

The main advantages of the direct multiple shooting over the direct single shooting can be attributed to the simultaneous integration on each time segment, the initialisation for state trajectory, and the improved convergence especially for nonlinear and unstable systems [21]. It is worth pointing out that, as discussed earlier, more optimisation variables and equality constraints result from the direct multiple shooting than from the direct single shooting. In other words, a larger nonlinear programming (NLP) problem is produced from the direct multiple shooting. Fortunately, the resulting larger but sparser NLP can be exploited by tools like IPOPT [46], which still makes the direct multiple shooting considerably faster. It is also interesting to note that the number of iterations per optimisation step resulting from the direct multiple shooting is fewer than that resulting from the direct single shooting, as shown in Fig. 10(f). This may also contribute to the reduced computational cost in the direct multiple shooting. In summary, with a proper optimisation technique, especially for a complicated system like the present one, MHE is still a favourable choice at the cost of slightly increased computational time.

The effects of horizon length and optimisation techniques have been ascertained, but discussions on the real-time implementability of MHE-type approaches are far from complete. The main driving force to deploy them in autonomous vehicles is to reduce the computational costs. Further investigation of algorithmic and technical simplification is of great importance, such as real-time iteration techniques, computation delay compensations, separation of preparation and feedback phases, and new code generation methods, as proposed in [21], [47], [48]. In addition, the computational burden of MHE-type approaches can also be considerably relieved or eliminated when there are richer computing resources within the considered vehicle or timely accesses to the cloud platform are available.

V. Conclusion

This paper has performed a comprehensive evaluation of MHE, mMHE, RLS-MHE and UKF for online estimating critical vehicle states in the presence of persistent noise, modelling error and external disturbance. This was based upon a highly nonlinear vehicle and tyre model that was justified to be locally observable. The accuracy, convergence, robustness and computational efficiency of these estimation algorithms were systematically compared under three driving manoeuvres and with different tuning parameters or optimisation techniques.

Illustrative results demonstrated that regarding the convergence rate and estimation accuracy, MHE had similar performance as mMHE but was clearly superior to UKF and RLS-MHE under all three manoeuvres. During robustness tests, mMHE could well mimic MHE in terms of compensating for deviated initialisation and modelling errors, but not in handling external disturbance. Meanwhile, MHE had better performance than UKF in all the three robustness tests. Under the circular manoeuvre where relatively large modelling uncertainties exist, mMHE suffered from increased estimation errors when considering longer horizons. When direct multiple shooting was employed, MHE was as computationally efficient as mMHE.

Future research will be concentrated on rigorous analysis of these estimation algorithms and exploring theoretical insights regarding the effects of different tuning parameters and operating conditions. Techniques to further improve the MHE computational efficiency will also be developed with experimental demonstration in autonomous vehicles.

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